Math 11

Quiz submitted

Cylinders & Quadric Surfaces (section 12.6)

- Due Sep 14 at 11:59pm
- Points 17
- Questions 17
- Available until Sep 15 at 11:59pm
- Time Limit None

Instructions

Math 110C	Cylinders & Quadric Surfaces (section 12.6)

This corresponds to section 12.6 in your Calculus textbook.

- Here are **notes** 📝 **before lesson. (https://ccsf.instructure.com/courses/68102/files/14363608?wrap=1)** ⬇ **(https://ccsf.instructure.com/courses/68102/files/14363608/download?download_frd=1)**
- Here is our lesson video:

Math 110C 12.6 Quadric Surfaces - a complete lesson

- Read section 12.6 of your Calculus textbook
- Check your

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Now do **Math 110C HW 12.6** on [www.WebAssign.net (http://www.WebAssign.net) . (http://www.webassign.net/)](http://www.webassign.net/)

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	9 minutes	17 out of 17

Score for this quiz: 17 out of 17

Submitted Sep 12 at 2:49am

This attempt took 9 minutes.

Correct answer

Question 1

1 / 1 pts

The curves of intersection of the surface with planes parallel to the coordinate planes are called traces of the surface.

Answer 1:

rulings

traces

Correct answer

Question 2

1 / 1 pts

A cylinder is a surface that consists of all lines that are parallel to a given line and pass through a given plane curve.

- (●) True
- ( ) False

Correct answer

Question 3

1 / 1 pts

The graph of the surface $z = x^2 z = x^2$ is a cylinder .

Answer 1:

cylinder

parabola

Correct answer

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Question 4

1 / 1 pts

The rulings of the surface $z = x^2 z = x^2$ are lines parallel to the y-axis.

- (selected) True
- False

Correct answer

Question 5

1 / 1 pts

The traces of the surface $z = x^2 z = x^2$ are parabolas.

- (selected) True
- False

Correct answer

Question 6

1 / 1 pts

In 3D, an equation like $x^2 + y^2 = 1 x^2 + y^2 = 1$ represents a cylinder .

Answer 1:

cylinder

circle

Correct answer

Question 7

1 / 1 pts

If one of the variables x, y, or z is missing from the equation of a surface, then the surface is a cylinder.

- (selected) True
- False

Correct answer

Question 8

1 / 1 pts

The trace of the surface $x^2 + y^2 = 1 x^2 + y^2 = 1$ in the xy-plane is a circle with equations $x^2 + y^2 = 1$ $x^2 + y^2 = 1$, $z = 0 z = 0$.

- (●) True
- ( ) False

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Correct answer

Question 9

1 / 1 pts

A quadric surface is the graph of a 2nd-degree equation in three variables x, y, and z.

- (●) True
- ( ) False

Correct answer

Question 10

1 / 1 pts

Quadric surfaces are the counterparts in 3D of the conic sections in the 2D plane.

- (●) True
- ( ) False

Correct answer

Question 11

1 / 1 pts

The graph of $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$ is an ellipsoid.

- (●) True
- ( ) False

Correct answer

Question 12

1 / 1 pts

The graph of $z = 4x^2 + y^2 z = 4x^2 + y^2$ is a cylinder.

- ( ) True
- (●) False

Correct answer

Question 13

1 / 1 pts

The graph of $z = y^2 - x^2 z = y^2 - x^2$ is a hyperbolic paraboloid.

- (●) True
- ( ) False

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Correct answer

Question 14

1 / 1 pts

The graph of $\frac{x^2}{4} + y^2 - \frac{z^2}{4} = 1 \frac{x^2}{4} + y^2 - \frac{z^2}{4} = 1$ is a hyperboloid of two sheets.

- ( ) True
- (●) False

Correct answer

Question 15

1 / 1 pts

The surface of the earth is an example of a quadric surface.

- (●) True
- ( ) False

Correct answer

Question 16

1 / 1 pts

Circular paraboloids, obtained by rotating a parabola about its axis, are used to collect and reflect light, sound, and radio and television signals.

- (●) True
- ( ) False

Correct answer

Question 17

1 / 1 pts

Cooling towers for nuclear reactors are designed in the shape of hyperboloids for reasons of structural stability.

Answer 1:

hyperboloids

paraboloids

ellipsoids

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